

Kavin Sriraj

+65 93737009 | srirajkavin@u.nus.edu | ka1525@ic.ac.uk | linkedin.com/in/kavin-sriraj
Singapore, 119278

EDUCATION

National University of Singapore	Singapore
Bachelor of Engineering (Electrical Engineering), Honors	Aug. 2023 – Present
Imperial College London	London, UK
Exchange Student, Electrical and Electronic Engineering	Sep. 2025 – June 2026

WORK EXPERIENCE

Software Intern	May 2025 – Aug. 2025
Kabam Robotics	Singapore
- Worked as a full-stack intern developing a web-based robotics dashboard, implementing frontend designs and backend functionality. - Led the integration of Mixpanel analytics to enable data-driven UI/UX design improvements, enhancing dashboard usability and user engagement.	
Research Engineering Intern	May 2024 – July 2024
A* Institute of High Performance Computing	Singapore
- Developed a comprehensive web application for visualizing urban fluid dynamic post-processing data, integrating both front-end and back-end components. - Implemented advanced parsing techniques for VTP files, enhancing data processing capabilities and efficiency. - Developed a full-stack application using Three.js for 3D visualizations, React (TypeScript) for a responsive frontend, and Django for robust backend processing, enabling interactive fluid dynamics simulation displays and efficient data management.	

BACHELOR THESIS

StereoGS: Stereo Video Reconstruction Using Dynamic Gaussian Splatting	
Imperial College London	Sep. 2025 – June 2026
- Reconstructed stereo video from monocular footage by fitting deformable 4D Gaussians to Structure-from-Motion camera poses and rendering synchronized left and right views from virtual stereo cameras. - Avoided monocular depth priors, adding optical-flow motion priors and a synthetic stereo renderer. - Evaluated on the South Kensington Stereo Video dataset with PSNR, SSIM, LPIPS, temporal flicker, TEPE and stereo disparity consistency.	

PUBLICATIONS

- Vijayalakshmi, K., Rajakannu, A., Ramachandran, K. P., & Sri Rajkavin, A. V. (2024). Condition monitoring of CNC drill bit for the manufacturing sector using wavelet analysis and artificial neural network based on feedforward multilayer perceptron (MLP). DOI: 10.14445/23488379/IJEEE-V11I12P106
- Vijayalakshmi, K., Rajakannu, A., Mohsina, K., Ramachandran, K. P., & Sri Rajkavin, A. V. (2025). Performance analysis of machine learning and deep learning techniques in diagnosing imbalance using machine fault simulator: A case study. DOI: 10.14445/22315381/IJETT-V73I1P123

PROJECTS

Tennis Match Prediction Model
Developed a machine learning model using XGBoost algorithm to predict tennis match outcomes based on player statistics and historical performance data.
Custom 4-DOF Robot Arm Kinematics
Developed forward and inverse kinematics algorithms from scratch for an OpenManipulator-X robot arm using MATLAB, utilizing pure math without relying on external robotics toolboxes.

TECHNICAL SKILLS

- Languages:** English, Tamil, Hindi
- Programming:** TypeScript, Python, C, C++, SQL, Rust, Julia, MATLAB, OCaml
- Frameworks:** React, Django, Three.js, React Three Fiber, PyTorch, XGBoost
- Tools:** Git, Linux, Mixpanel, KiCad, STM32CubeIDE, Fusion360